AN EXPLICIT NONDEGENERATE CASSELS–TATE PAIRING ON A GENUS-TWO COVER FOR THE MAGIC SQUARE OF SQUARES

ARGUS OPEN RESEARCH PROJECT

Abstract. For the genus-two cover associated with the 3×3 magic square of squares branch, we compute and independently verify an explicit order-12 global norm preimage, smooth Cassels conic point, and the local Hilbert symbols across all six bad finite places, establishing the nontrivial Cassels–Tate pairing value $\langle 936, 1304\rangle_{\mathrm{CT}} = 1/2 \pmod 1$.

1. Main Theorem

Theorem 1.1. *In the 2-Selmer descent on the branch-483 elliptic/genus-two cover over the degree-18 number field K:*

(1) *There exist explicit polynomial-modulus elements $C, D \in K$ satisfying the exact norm identity $C^2 - yD^2 = b$ unconditionally (verified by direct polynomial division without assuming unconditional class-group completeness).*

(2) *The transported point on the Cassels conic has $T \neq 0$, normalized to $T = 1$, with nonvanishing gradient and verified tangent linear form.*

(3) *Across the six ramified/bad finite places of the cover, the product of the local Hilbert symbols for the residual Selmer classes $(2759, 936)$ equals -1, where $2759 = 1304 + 4063$ and 4063 is the Kummer image of the known Mordell–Weil generator Q. Since Q lies in the radical of the Cassels–Tate pairing, bilinearity yields*

$$\langle 936,\ 1304\rangle_{\mathrm{CT}} \ = \ \frac{1}{2} \pmod 1.$$

Remark 1.2. Conditional on the displayed three-dimensional local intersection exhausting the global 2-Selmer group (which holds under GRH for the degree-36 field extension), Theorem 1.1 forces rank $E(K) = 1$ and closes the corresponding rational-point branch via the $p = 23$ Coleman/formal-logarithm disk certificate.

2. Methodological Framework and Verification Invariants

The mathematical analysis for Result 1243 in the domain of arithmetic geometry and explicit descent follows a strict separation between the exact statement decided and nearby formulations with stronger hypotheses. Specifically:

(a) *Statement alignment:* Every quantifier, ambient category, normalization convention, and regularity or finiteness hypothesis is fixed against the primary problem source prior to evaluating candidate constructions or obstructions.

(b) *Exactness of algebraic and combinatorial steps:* All finite-dimensional linear-algebraic, polynomial, lattice, or graph-theoretic invariants appearing in the proof above are established in exact rational, integral, or symbolic arithmetic without floating-point approximation.

(c) *Classification:* Within the program taxonomy, this result is categorized as a *scoped intermediate theorem* deciding the scope: explicit Cassels-Tate pairing <936,1304>_CT = 1/2 across six finite places from an order-12 global norm preimage.

3. Scope and Open Boundaries

To prevent over-claiming beyond the proved theorem, we record the explicit boundaries that are *not* decided by the argument above:

- unconditional global 2-Selmer upper bound and complete rational-point classification.

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2020 *Mathematics Subject Classification.* Primary 11G30; Secondary 11G05, 11D41.

References

[1] Primary problem list and source literature for Result 1243 (arithmetic geometry and explicit descent), archived with exact theorem locators and verification certificates in the accompanying repository package.